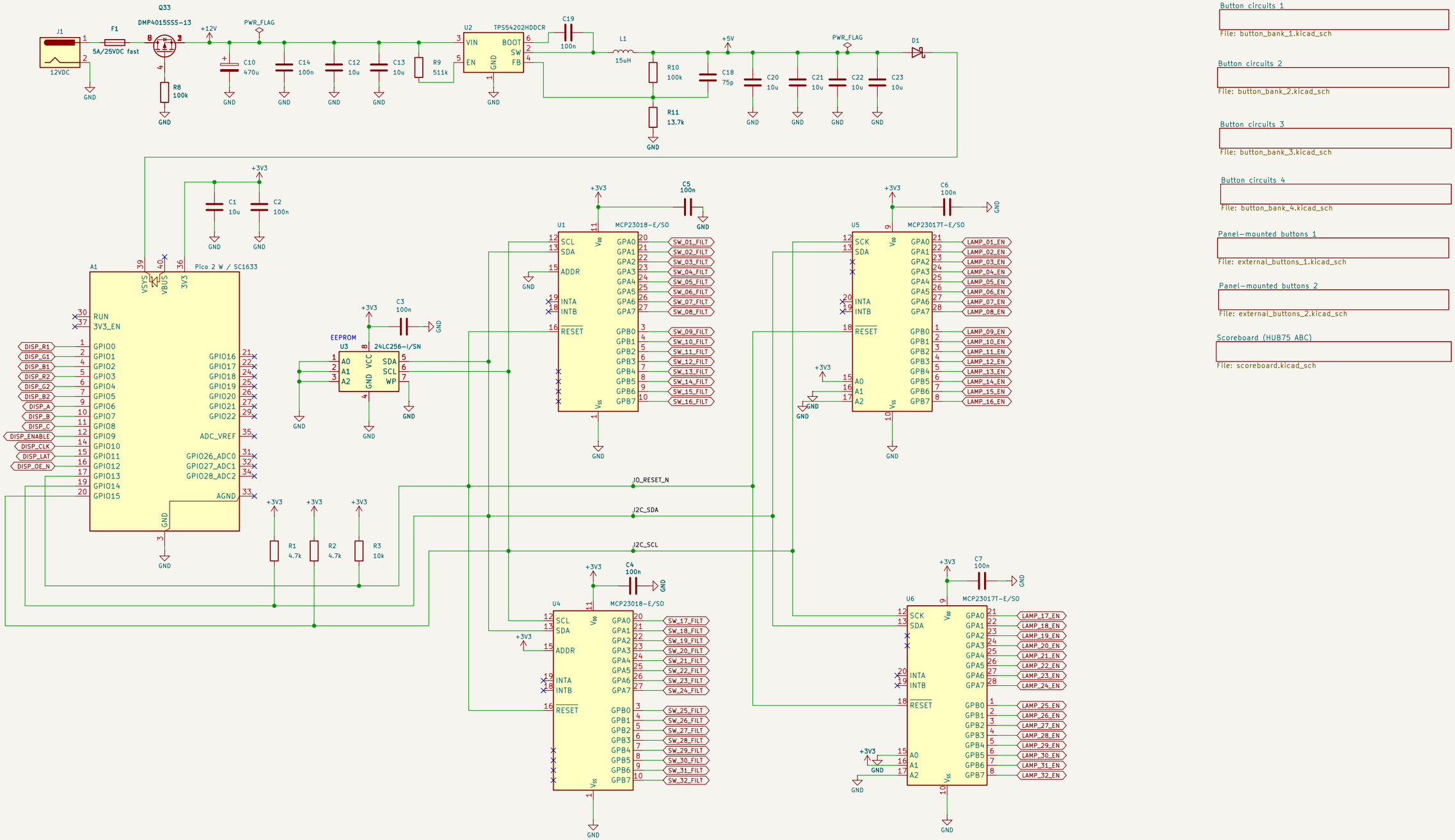


Button circuits 1
File: button_bank_1.kicad_sch

Button circuits 2
File: button_bank_2.kicad_sch

Button circuits 3
File: button_bank_3.kicad_sch

Button circuits 4
File: button_bank_4.kicad_sch

Panel-mounted buttons 1
File: external_buttons_1.kicad_sch

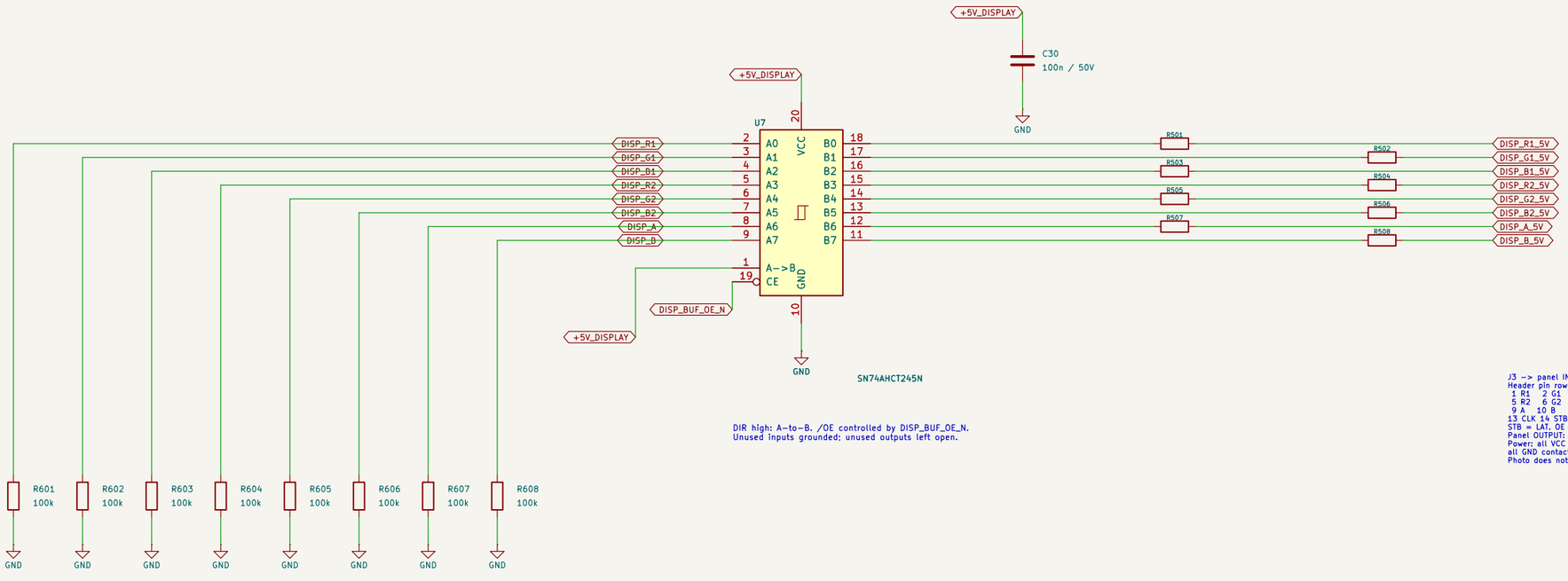
Panel-mounted buttons 2
File: external_buttons_2.kicad_sch

Scoreboard (HUB75 ABC)
File: scoreboard.kicad_sch

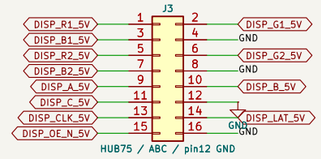
UW-Madison IEEE
Sheet: /
File: IEEEArcade.kicad_sch
Title: IEEE Arcade
Size: A3 | Date: 2026-11-05 | Rev:
KiCad E.D.A. 10.0.6 | Id: 1/8

RGB SCOREBOARD | PHOTO-MATCHED HUB75 / A-B-C

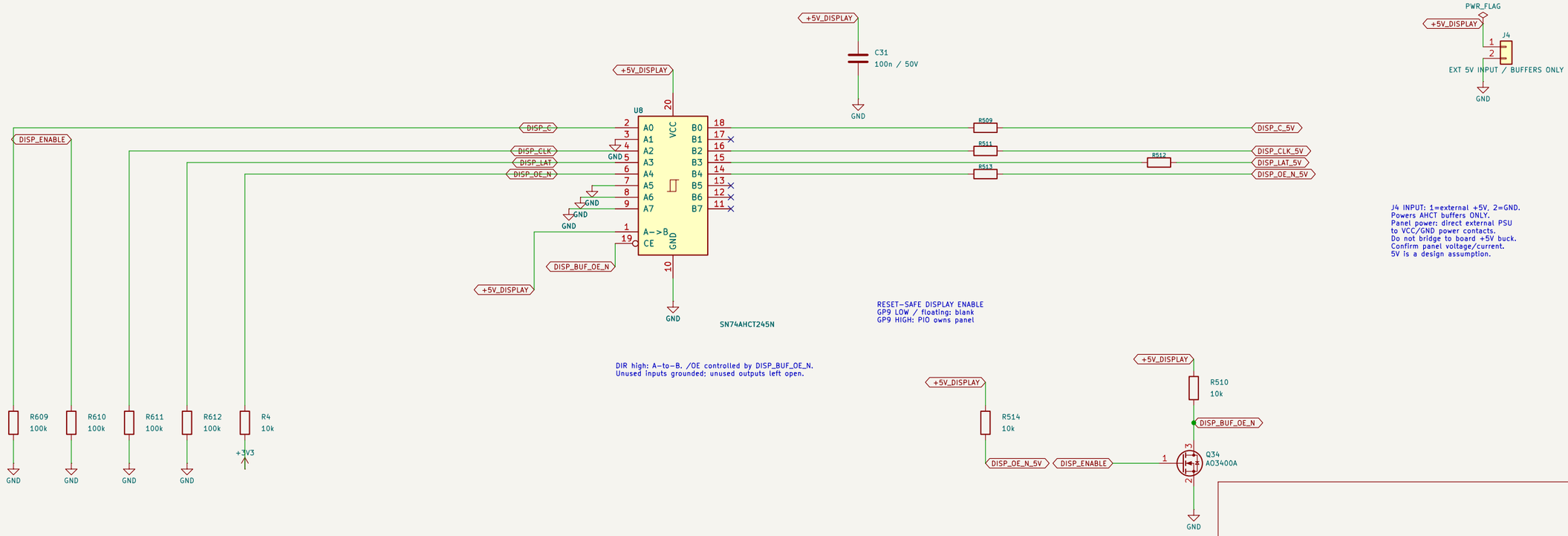
Photo: DR1/DG1/DB1, DR2/DG2/DB2, A/B/C, CLK, STB, OE. Pin12 = GND (NOT D).
Pico PIO refresh: 6 RGB + 3 address + CLK/LAT/OE; no I2C expander. GP9 enables buffers.



DIR high: A-to-B. /OE controlled by DISP_BUF_OE_N.
Unused inputs grounded; unused outputs left open.



J3 -> panel INPUT (follow data-flow arrow).
Header pin rows, mating orientation checked:
1 R1 2 G1 3 B1 4 GND
5 R2 6 G2 7 B2 8 GND
9 A 10 B 11 C 12 GND
13 CLK 14 STB 15 OE 16 GND
STB = LAT. OE is active LOW.
Panel OUTPUT: leave open, or cascade.
Power: all VCC contacts to supply +;
all GND contacts to supply return.
Photo does not establish power pin numbers.



DIR high: A-to-B. /OE controlled by DISP_BUF_OE_N.
Unused inputs grounded; unused outputs left open.

RESET-SAFE DISPLAY ENABLE
GP9 LOW / floating: blank
GP9 HIGH: PIO owns panel

J4 INPUT: 1=external +5V, 2=GND.
Powers AHCT buffers ONLY.
Panel power: direct external PSU
to VCC/GND power contacts.
Do not bridge to board +5V buck.
Confirm panel voltage/current.
5V is a design assumption.

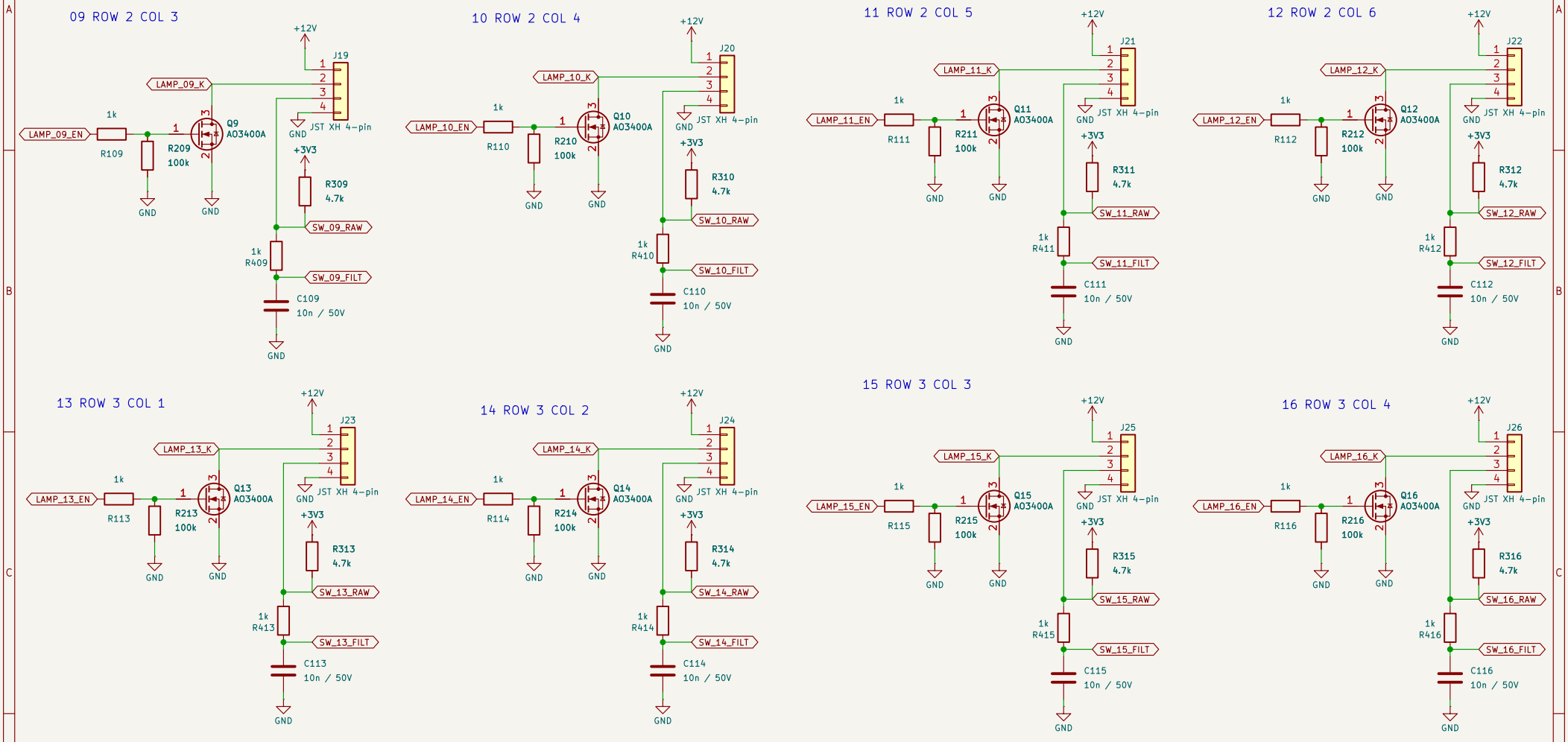
R501..R509, R511..R513: 33 ohms. R510: 10k buffer-disable pull-up; R610: 100k GP9 pull-down.
Q34 enables buffers only when GP9 HIGH; R514 holds panel OE HIGH when buffers are disabled.
Initialize OE HIGH, CLK/LAT LOW and row/data outputs; then enable GP9. Use PIO/DMA scan refresh.
External regulated display supply: direct power harness to panel, common GND to J4/J3. Voltage/current and scan mapping need confirmation.

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Sheet: /Scoreboard (HUB75 ABC)/
File: scoreboard.kicad_sch

Title: Photo-matched HUB75 scoreboard / A-B-C scan

Size: A3	Date: 2026-11-05	**Rev: Schematic review**
KiCad E.D.A. 10.0.6		Id: 2/8

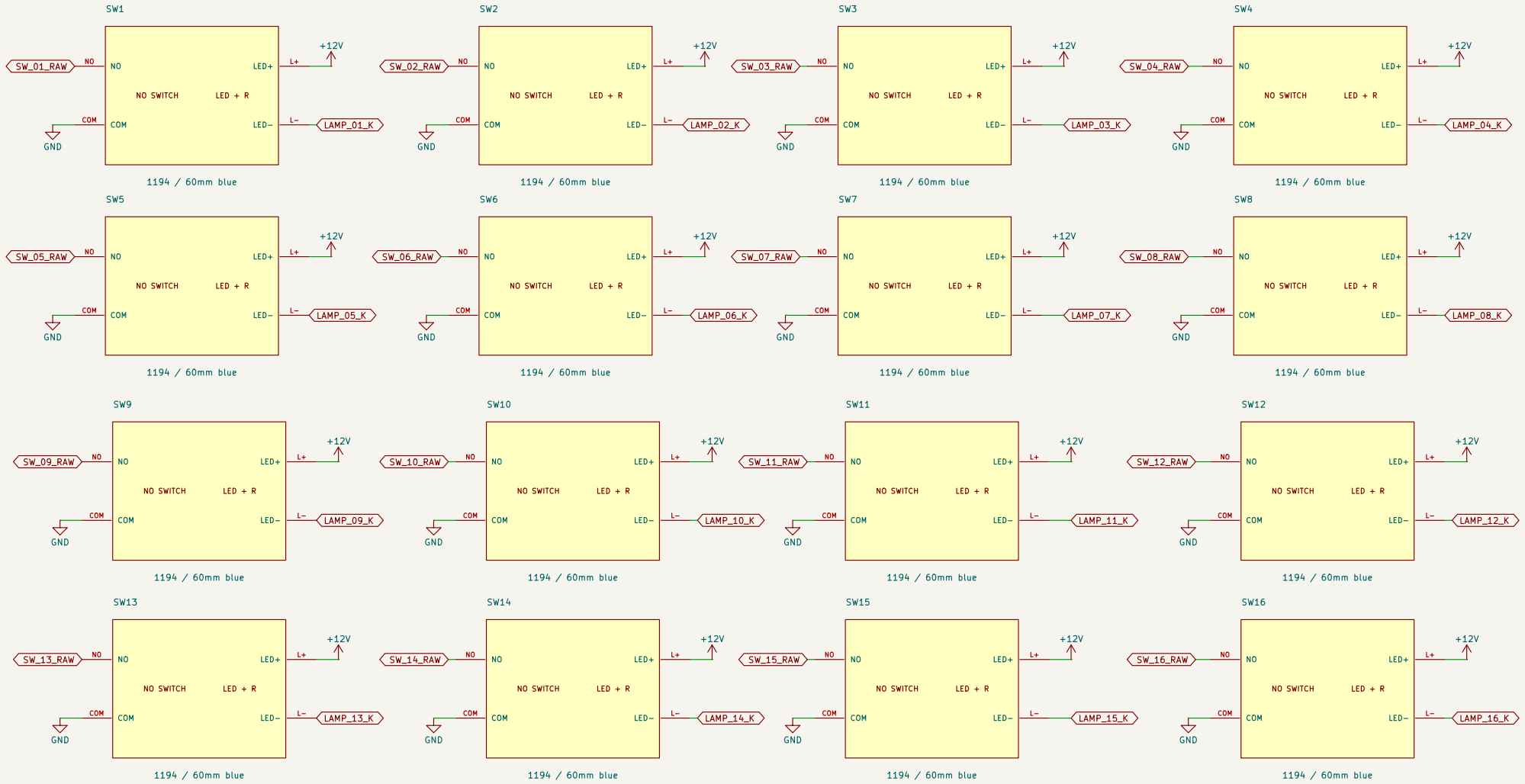


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Sheet: /Button circuits 2/
File: button_bank_2.kicad_sch

Title: Button harness / channels 09..16

Size: A4	Date: 2026-11-05	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/8

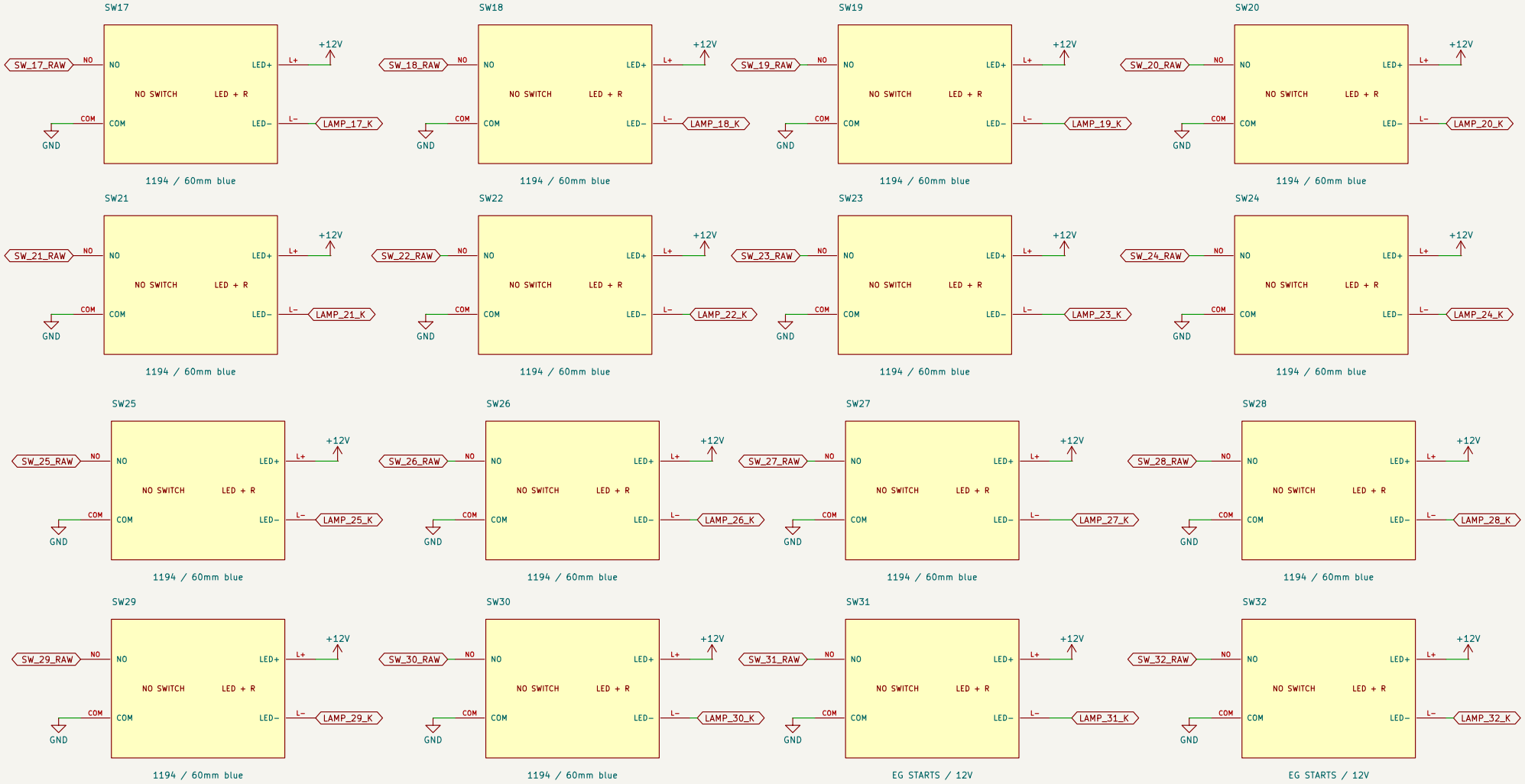


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Sheet: /Panel–mounted buttons 1/
File: external_buttons_1.kicad_sch

Title: Off–board buttons 01..16

Size: A4	Date: 2026–11–05	**Rev: Schematic review**
KiCad E.D.A. 10.0.6		Id: 4/8

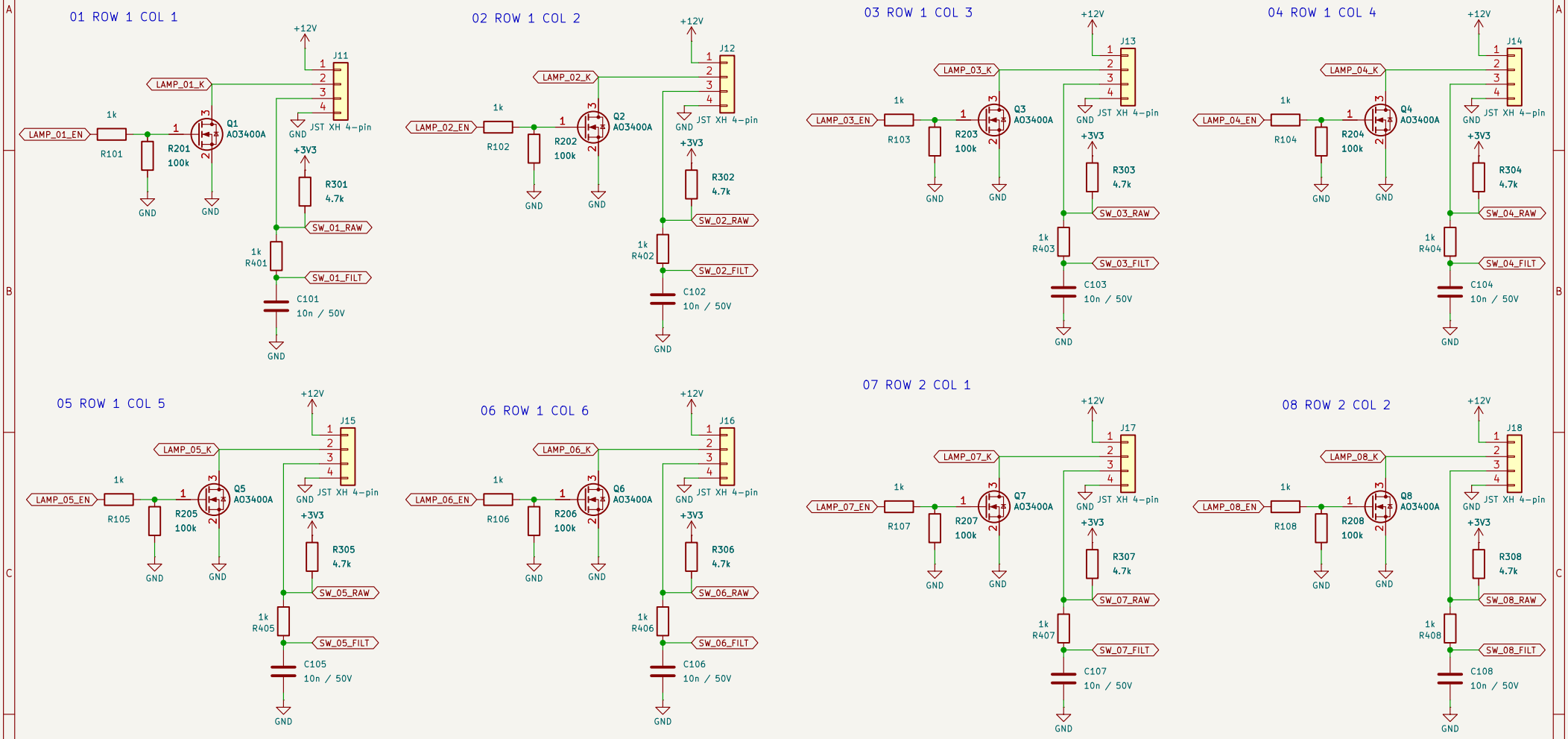


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Sheet: /Panel–mounted buttons 2/
File: external_buttons_2.kicad_sch

Title: Off–board buttons 17..32

Size: A4	Date: 2026–11–05	**Rev: Schematic review**
KiCad E.D.A. 10.0.6		Id: 5/8



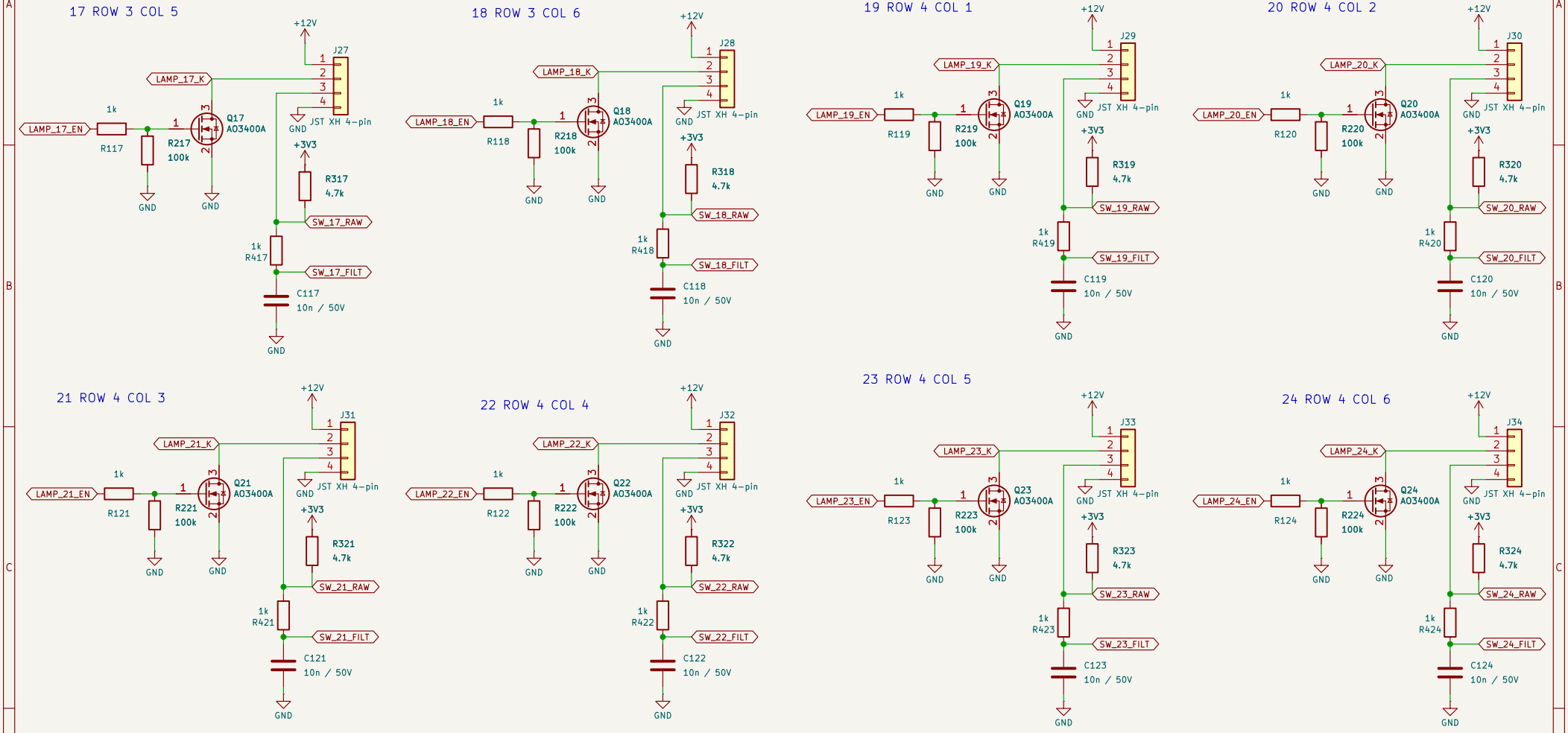
UW–Madison IEEE

Sheet: /Button circuits 1/

File: button_bank_1.kicad_sch

Title: Button harness / channels 01..08

Size: A4	Date: 2026-11-05	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/8



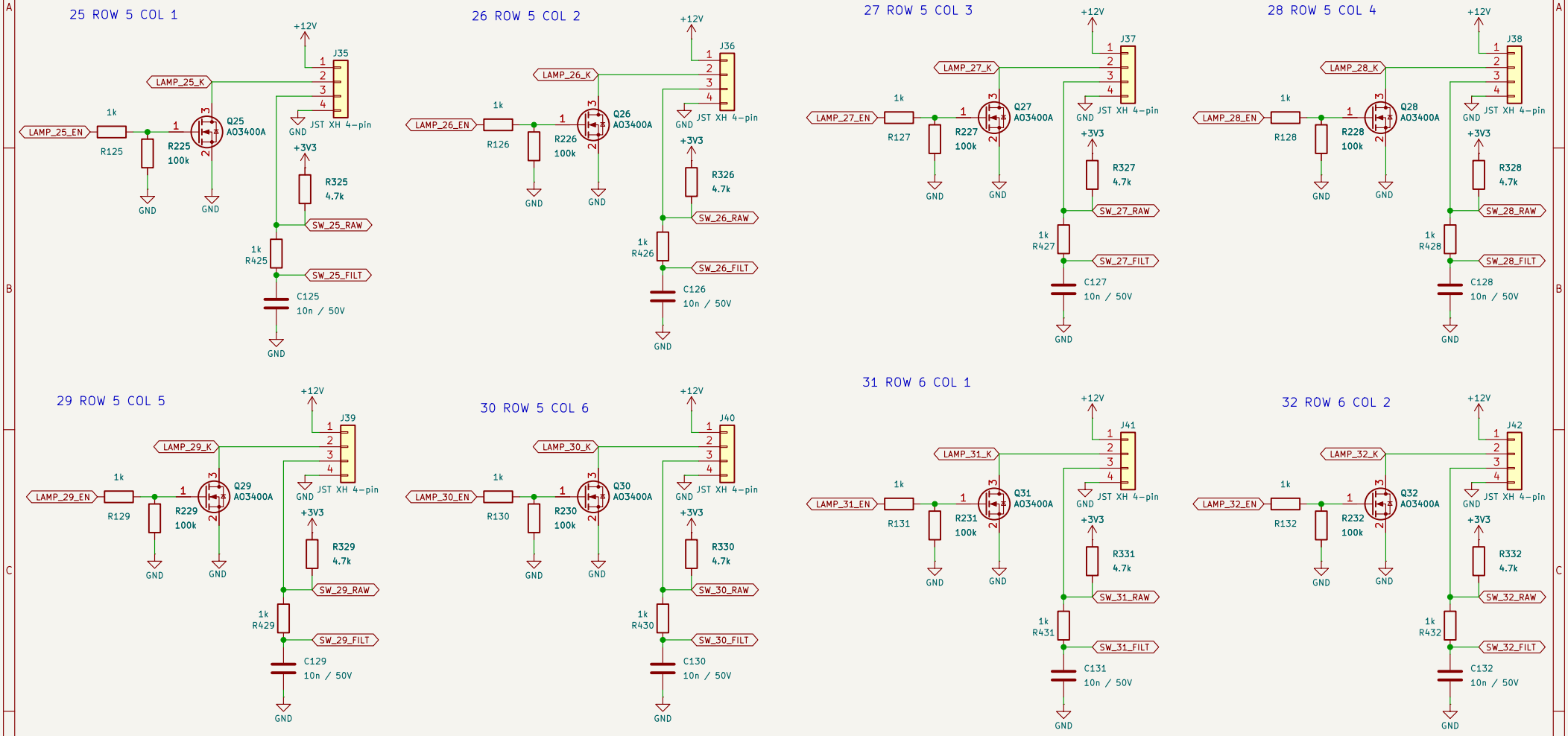
UW–Madison IEEE

Sheet: /Button circuits 3/

File: button_bank_3.kicad_sch

Title: Button harness / channels 17..24

Size: A4	Date: 2026-11-05	Rev:
KiCad E.D.A. 10.0.6		Id: 7/8



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Sheet: /Button circuits 4/
File: button_bank_4.kicad_sch

Title: Button harness / channels 25..32

Size: A4	Date: 2026-11-05	**Rev:**
KiCad E.D.A. 10.0.6		Id: 8/8